

Uncovering_Hidden_Country_Segments_for_Global_Development_Strategy

June 16, 2026

```
[2]: import numpy as np
import pandas as pd
```

```
[3]: !gdown 1SWUXaC78y2lnahVnUwBPmyX0icJW41Po
```

Downloading...

From: <https://drive.google.com/uc?id=1SWUXaC78y2lnahVnUwBPmyX0icJW41Po>

To: /content/Country-data.csv

0% 0.00/9.23k [00:00<?, ?B/s] 100% 9.23k/9.23k [00:00<00:00, 20.3MB/s]

```
[4]: country_data = pd.read_csv('Country-data.csv')

country_data.head()
```

```
[4]:
```

	country	child_mort	exports	health	imports	income	\
0	Afghanistan	90.2	10.0	7.58	44.9	1610	
1	Albania	16.6	28.0	6.55	48.6	9930	
2	Algeria	27.3	38.4	4.17	31.4	12900	
3	Angola	119.0	62.3	2.85	42.9	5900	
4	Antigua and Barbuda	10.3	45.5	6.03	58.9	19100	

	inflation	life_expec	total_fer	gdpp
0	9.44	56.2	5.82	553
1	4.49	76.3	1.65	4090
2	16.10	76.5	2.89	4460
3	22.40	60.1	6.16	3530
4	1.44	76.8	2.13	12200

```
[5]: import matplotlib.pyplot as plt
import seaborn as sns

# Check for missing values and data types
print("--- Data Info ---")
display(country_data.info())

# Statistical summary
print("\n--- Statistical Summary ---")
```

```
display(country_data.describe())
```

--- Data Info ---

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 167 entries, 0 to 166

Data columns (total 10 columns):

#	Column	Non-Null Count	Dtype
0	country	167 non-null	object
1	child_mort	167 non-null	float64
2	exports	167 non-null	float64
3	health	167 non-null	float64
4	imports	167 non-null	float64
5	income	167 non-null	int64
6	inflation	167 non-null	float64
7	life_expec	167 non-null	float64
8	total_fer	167 non-null	float64
9	gdpp	167 non-null	int64

dtypes: float64(7), int64(2), object(1)

memory usage: 13.2+ KB

None

--- Statistical Summary ---

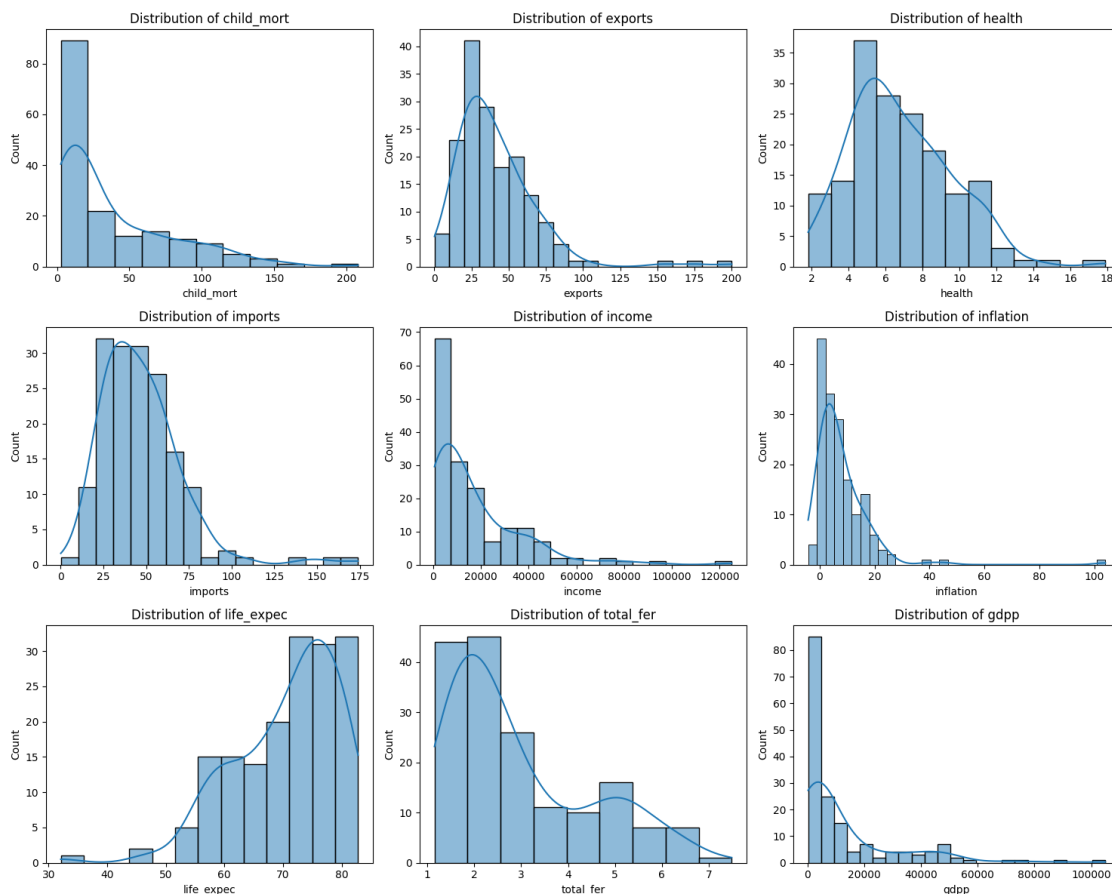
	child_mort	exports	health	imports	income \
count	167.000000	167.000000	167.000000	167.000000	167.000000
mean	38.270060	41.108976	6.815689	46.890215	17144.688623
std	40.328931	27.412010	2.746837	24.209589	19278.067698
min	2.600000	0.109000	1.810000	0.065900	609.000000
25%	8.250000	23.800000	4.920000	30.200000	3355.000000
50%	19.300000	35.000000	6.320000	43.300000	9960.000000
75%	62.100000	51.350000	8.600000	58.750000	22800.000000
max	208.000000	200.000000	17.900000	174.000000	125000.000000

	inflation	life_expec	total_fer	gdpp
count	167.000000	167.000000	167.000000	167.000000
mean	7.781832	70.555689	2.947964	12964.155689
std	10.570704	8.893172	1.513848	18328.704809
min	-4.210000	32.100000	1.150000	231.000000
25%	1.810000	65.300000	1.795000	1330.000000
50%	5.390000	73.100000	2.410000	4660.000000
75%	10.750000	76.800000	3.880000	14050.000000
max	104.000000	82.800000	7.490000	105000.000000

0.0.1 Data Preprocessing & EDA

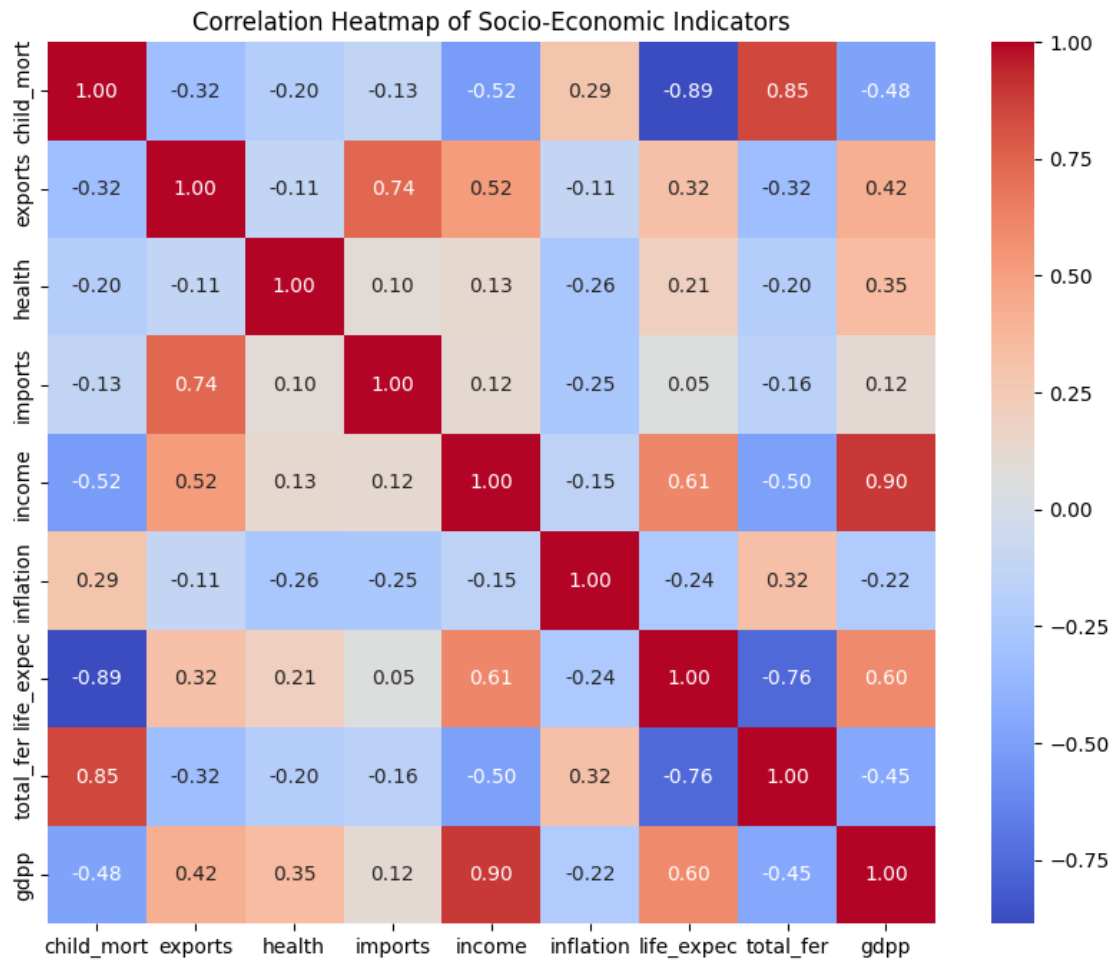
```
[6]: # Visualize distributions of key features
features = ['child_mort', 'exports', 'health', 'imports', 'income', 'inflation', 'life_expec', 'total_fer', 'gdpp']

plt.figure(figsize=(15, 12))
for i, col in enumerate(features):
    plt.subplot(3, 3, i + 1)
    sns.histplot(country_data[col], kde=True)
    plt.title(f'Distribution of {col}')
plt.tight_layout()
plt.show()
```



```
[7]: # Correlation heatmap to identify relationships
plt.figure(figsize=(10, 8))
sns.heatmap(country_data[features].corr(), annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Heatmap of Socio-Economic Indicators')
```

```
plt.show()
```



0.0.2 Feature Scaling

```
[8]: # Outlier Treatment: Using capping (percentile) method to handle extreme values
# This helps in preventing clusters from being dominated by extreme outliers

country_data_clean = country_data.copy()
for col in features:
    q1 = country_data_clean[col].quantile(0.05)
    q3 = country_data_clean[col].quantile(0.95)
    country_data_clean[col] = country_data_clean[col].clip(lower=q1, upper=q3)

print("Outliers have been capped at 5th and 95th percentiles.")
```

Outliers have been capped at 5th and 95th percentiles.

/tmp/ipykernel_2130/887244674.py:8: FutureWarning: Downcasting behavior in

Series and DataFrame methods 'where', 'mask', and 'clip' is deprecated. In a future version this will not infer object dtypes or cast all-round floats to integers. Instead call `result.infer_objects(copy=False)` for object inference, or cast round floats explicitly. To opt-in to the future behavior, set `pd.set_option('future.no_silent_downcasting', True)`

```
country_data_clean[col] = country_data_clean[col].clip(lower=q1, upper=q3)
```

```
[9]: from sklearn.preprocessing import StandardScaler

# Scaling the data
scaler = StandardScaler()
data_scaled = scaler.fit_transform(country_data_clean[features])

# Convert to DataFrame for easier inspection
country_data_scaled = pd.DataFrame(data_scaled, columns=features)
print("Data scaling complete. Here is the head of the scaled data:")
display(country_data_scaled.head())
```

Data scaling complete. Here is the head of the scaled data:

	child_mort	exports	health	imports	income	inflation	life_expec	\
0	1.479588	-1.371444	0.322232	-0.025440	-0.960575	0.387667	-1.825310	
1	-0.560024	-0.570493	-0.086218	0.177314	-0.395590	-0.404004	0.682454	
2	-0.263504	-0.045941	-1.030017	-0.765221	-0.193907	1.452825	0.707406	
3	2.194560	1.159521	-1.553469	-0.135037	-0.669255	2.215708	-1.338729	
4	-0.734610	0.312167	-0.292427	0.741740	0.227115	-0.891802	0.744836	

	total_fer	gdpp
0	2.020718	-0.757874
1	-0.887331	-0.523775
2	-0.022587	-0.499286
3	2.049310	-0.560839
4	-0.552591	0.012991

0.0.3 Clustering with K-Means

K-means clustering on just scaled data

```
[10]: from sklearn.cluster import KMeans

# Based on the plots, let's proceed with an optimal number of clusters (e.g., 3)
kmeans = KMeans(n_clusters=3, max_iter=50, random_state=42)
kmeans.fit(data_scaled)

# Assigning cluster labels back to the original data
country_data['cluster_id'] = kmeans.labels_
print("K-Means clustering with k=3 completed.")
display(country_data.groupby('cluster_id')[features].mean())
```

K-Means clustering with k=3 completed.

	child_mort	exports	health	imports	income \
cluster_id					
0	90.035294	27.790176	6.225686	41.399331	3873.000000
1	5.237838	58.097297	8.782973	51.281081	45056.756757
2	20.322785	41.750633	6.275190	48.378481	12639.746835

	inflation	life_expec	total_fer	gdpp
cluster_id				
0	11.932255	59.856863	4.833529	1843.529412
1	2.588432	79.956757	1.755676	42102.702703
2	7.534797	73.059494	2.289114	6496.126582

Determining optimal clusters using Elbow and Silhouette methods (scaled data)

```
[11]: from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

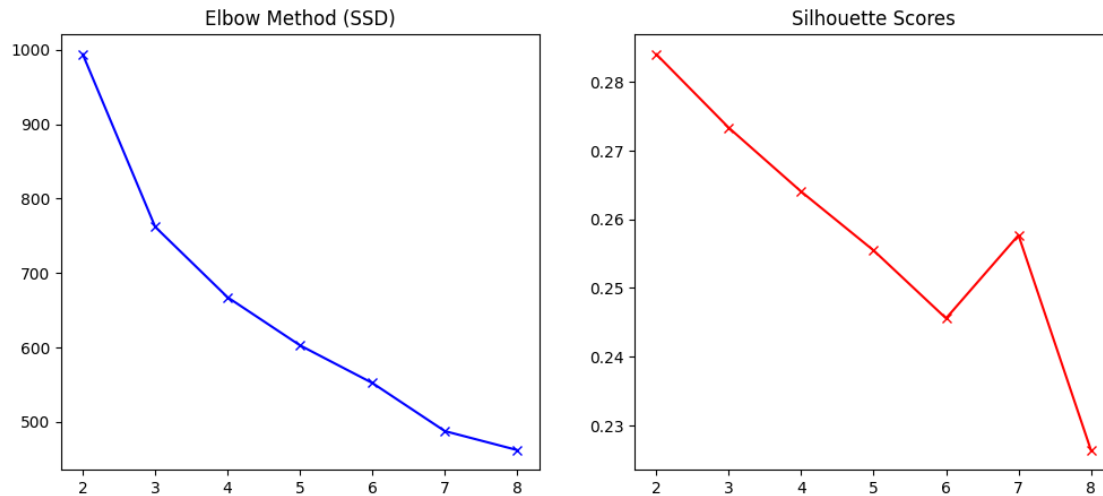
# Determining optimal clusters using Elbow and Silhouette methods
ssd = []
silhouette_scores = []
range_n_clusters = [2, 3, 4, 5, 6, 7, 8]

for num_clusters in range_n_clusters:
    kmeans = KMeans(n_clusters=num_clusters, max_iter=50, random_state=42)
    kmeans.fit(data_scaled)
    ssd.append(kmeans.inertia_)

    cluster_labels = kmeans.labels_
    silhouette_avg = silhouette_score(data_scaled, cluster_labels)
    silhouette_scores.append(silhouette_avg)

# Plotting Elbow and Silhouette
plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.plot(range_n_clusters, ssd, 'bx-')
plt.title('Elbow Method (SSD)')

plt.subplot(1, 2, 2)
plt.plot(range_n_clusters, silhouette_scores, 'rx-')
plt.title('Silhouette Scores')
plt.show()
```



```
[12]: import matplotlib.pyplot as plt
import seaborn as sns

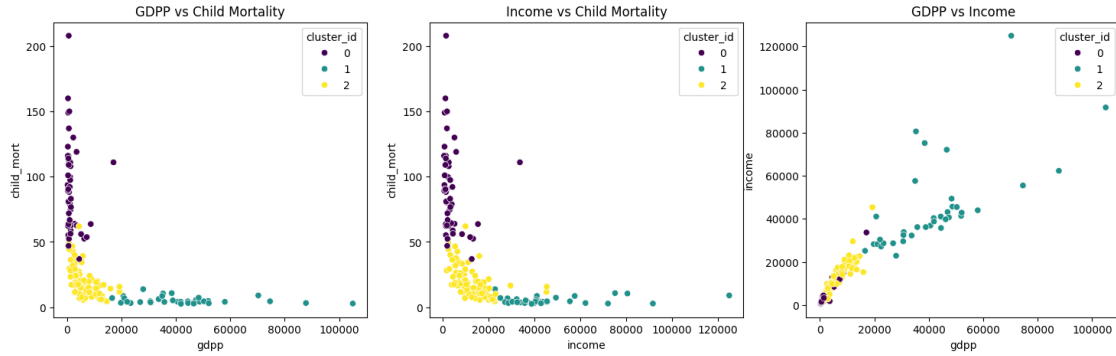
# Visualizing the clusters using key socio-economic metrics
plt.figure(figsize=(18, 5))

plt.subplot(1, 3, 1)
sns.scatterplot(data=country_data, x='gdp', y='child_mort', hue='cluster_id',
               ↪palette='viridis')
plt.title('GDP vs Child Mortality')

plt.subplot(1, 3, 2)
sns.scatterplot(data=country_data, x='income', y='child_mort',
               ↪hue='cluster_id', palette='viridis')
plt.title('Income vs Child Mortality')

plt.subplot(1, 3, 3)
sns.scatterplot(data=country_data, x='gdp', y='income', hue='cluster_id',
               ↪palette='viridis')
plt.title('GDP vs Income')

plt.show()
```



```
[13]: # Filtering countries in Cluster 0 (The high-priority group)
priority_countries = country_data[country_data['cluster_id'] == 0].
    ↪sort_values(by=['gdpp', 'income', 'child_mort'], ascending=[True, True,
    ↪False])

print(f'Number of countries identified in the priority cluster:
    ↪{len(priority_countries)}')
print('\nTop 10 countries in dire need of development funds:')
display(priority_countries[['country', 'child_mort', 'income', 'gdpp']].
    ↪head(10))
```

Number of countries identified in the priority cluster: 51

Top 10 countries in dire need of development funds:

	country	child_mort	income	gdpp
26	Burundi	93.6	764	231
88	Liberia	89.3	700	327
37	Congo, Dem. Rep.	116.0	609	334
112	Niger	123.0	814	348
132	Sierra Leone	160.0	1220	399
93	Madagascar	62.2	1390	413
106	Mozambique	101.0	918	419
31	Central African Republic	149.0	888	446
94	Malawi	90.5	1030	459
50	Eritrea	55.2	1420	482

```
[14]: # Filtering countries in Cluster 1 (The highly developed group)
cluster_1_countries = country_data[country_data['cluster_id'] == 1].
    ↪sort_values(by=['gdpp', 'income'], ascending=[False, False])

print(f"Number of countries identified in Cluster 1:
    ↪{len(cluster_1_countries)}")
print("\nTop 10 countries in Cluster 1 (sorted by highest GDP and Income):")
```



```
display(cluster_1_countries[['country', 'child_mort', 'income', 'gdpp', 'life_expec']].head(10))
```

Number of countries identified in Cluster 1: 37

Top 10 countries in Cluster 1 (sorted by highest GDP and Income):

	country	child_mort	income	gdpp	life_expec
91	Luxembourg	2.8	91700	105000	81.3
114	Norway	3.2	62300	87800	81.0
145	Switzerland	4.5	55500	74600	82.2
123	Qatar	9.0	125000	70300	79.5
44	Denmark	4.1	44000	58000	79.5
144	Sweden	3.0	42900	52100	81.5
7	Australia	4.8	41400	51900	82.0
110	Netherlands	4.5	45500	50300	80.7
73	Ireland	4.2	45700	48700	80.4
159	United States	7.3	49400	48400	78.7

0.0.4 Dimensionality Reduction (PCA)

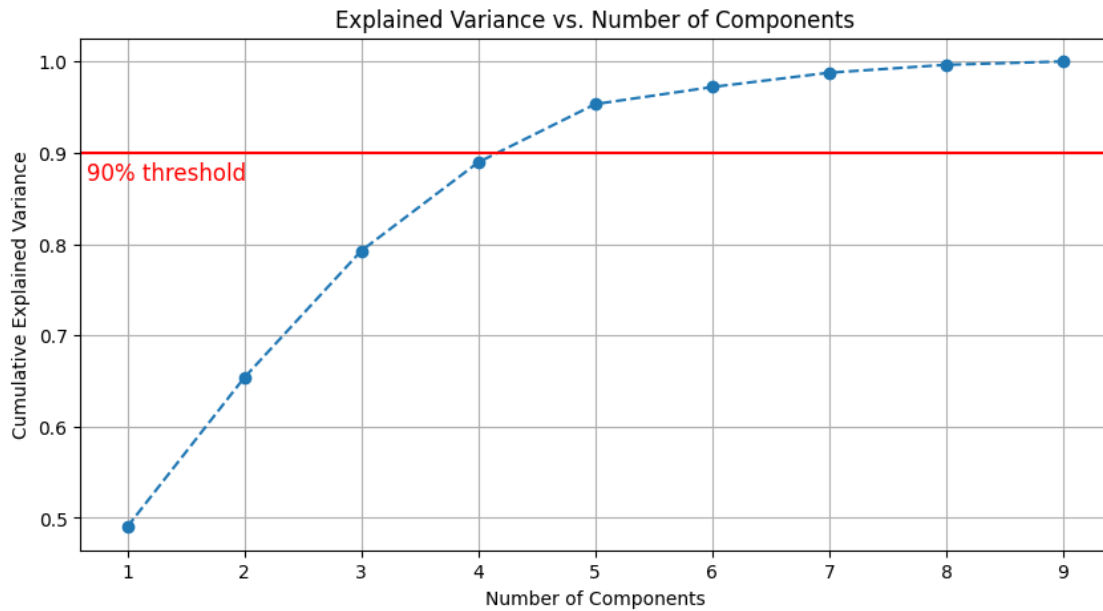
```
[15]: from sklearn.decomposition import PCA

# Applying PCA
pca = PCA(random_state=42)
pca.fit(data_scaled)

# Calculating Cumulative Variance
var_cum = np.cumsum(pca.explained_variance_ratio_)

# Plotting Explained Variance
plt.figure(figsize=(10, 5))
plt.plot(range(1, len(var_cum) + 1), var_cum, marker='o', linestyle='--')
plt.title('Explained Variance vs. Number of Components')
plt.xlabel('Number of Components')
plt.ylabel('Cumulative Explained Variance')
plt.axhline(y=0.90, color='r', linestyle='-')
plt.text(0.65, 0.87, '90% threshold', color = 'red', fontsize = 12)
plt.grid()
plt.show()

print(f"Cumulative variance explained by first 4 components: {var_cum[3]:.2f}")
```

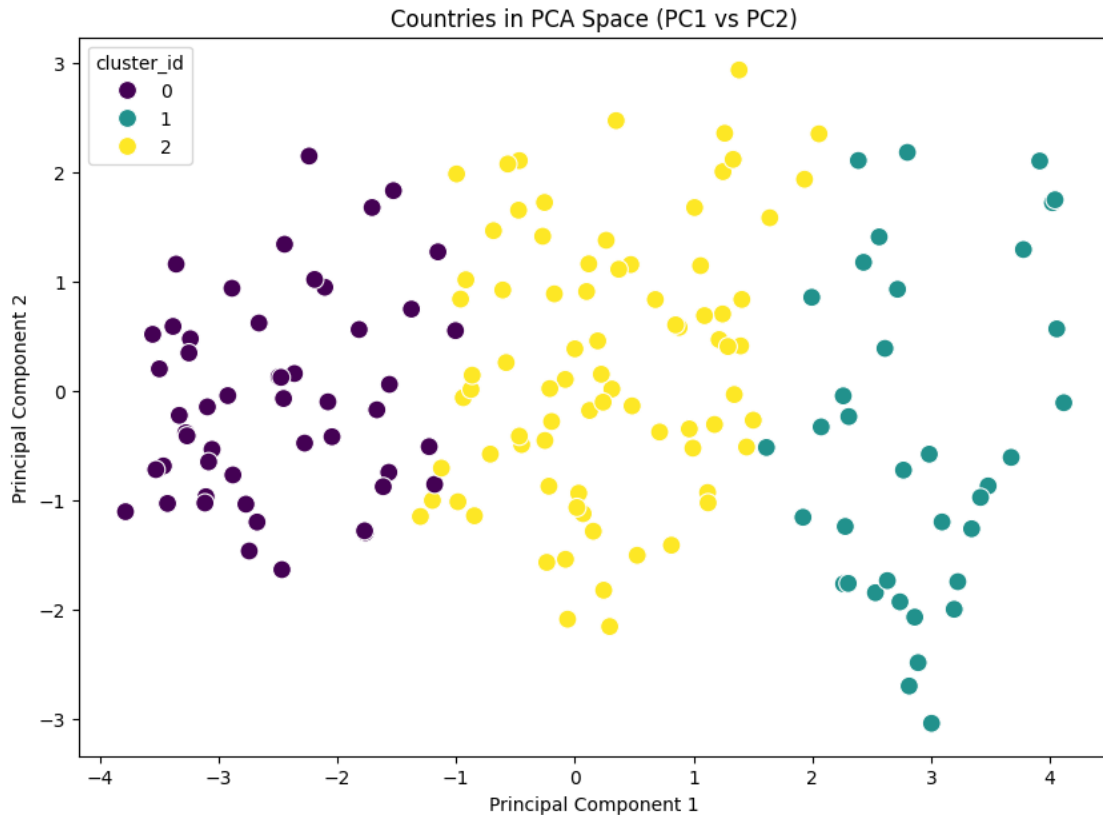


Cumulative variance explained by first 4 components: 0.89

```
[16]: # Using 4 components as they explain ~89% of the variance
pca_final = PCA(n_components=4, random_state=42)
data_pca = pca_final.fit_transform(data_scaled)

# Create a dataframe for PCA components
df_pca = pd.DataFrame(data_pca, columns=['PC1', 'PC2', 'PC3', 'PC4'])
df_pca['cluster_id'] = country_data['cluster_id']
df_pca['country'] = country_data['country']

# Visualizing PC1 vs PC2
plt.figure(figsize=(10, 7))
sns.scatterplot(x='PC1', y='PC2', hue='cluster_id', data=df_pca,
               palette='viridis', s=100)
plt.title('Countries in PCA Space (PC1 vs PC2)')
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.show()
```



```
[17]: # Examining Component Loadings
loadings = pd.DataFrame(pca_final.components_.T, columns=['PC1', 'PC2', 'PC3', 'PC4'], index=features)
print("Principal Component Loadings (weights of original features):")
display(loadings)
```

Principal Component Loadings (weights of original features):

	PC1	PC2	PC3	PC4
child_mort	-0.421623	0.032585	0.094262	0.387864
exports	0.232156	0.599838	-0.295282	0.252625
health	0.181381	-0.191686	0.646498	0.283765
imports	0.094524	0.729981	0.283589	0.012585
income	0.410887	-0.110986	-0.198015	0.419843
inflation	-0.225176	-0.079882	-0.590478	0.169649
life_expec	0.431881	-0.129883	-0.113004	-0.225216
total_fer	-0.406693	0.008781	0.060062	0.446607
gdpp	0.393765	-0.184042	-0.035916	0.499832

0.0.5 Dimensionality Reduction: PCA Observations & Summary

1. Component Selection

- The Scree Plot (Explained Variance vs. Number of Components) shows that the first **4 principal components** capture approximately **89% of the total variance** in the dataset.
- By retaining these 4 components, we successfully reduce the dimensionality from 9 features to 4 while preserving the vast majority of the socio-economic signals required for clustering.

2. Interpreting the Principal Components (Loadings)

- **PC1 (The Development Axis):** High positive loadings for `income`, `gdpp`, and `life_expec`, and a strong negative loading for `child_mort`. This component effectively measures a country's overall level of economic development and health status.
- **PC2 (The Trade Axis):** High positive loadings for `exports` and `imports`. This component differentiates countries based on their level of international trade activity regardless of their wealth.
- **PC3:** Is heavily influenced by `inflation` (negative) and `health` (positive) spending percentage.

3. Visualization in PCA Space

- The scatter plot of **PC1 vs PC2** reveals clear separation between the K-Means clusters.
- **Cluster 0 (High Priority)** is distinctly grouped at the negative end of **PC1**, confirming that their primary differentiator is low economic performance and poor health indicators.
- **Cluster 1 (Developed)** is located at the positive far-right of **PC1**.
- This visualization validates that our clustering algorithm captured the underlying socio-economic structure of the data revealed by PCA.

0.0.6 Clustering on PCA Components

K-means clustering on PCA data

```
[18]: # Performing K-Means on the 4 PCA components
kmeans_pca = KMeans(n_clusters=3, max_iter=50, random_state=42)
kmeans_pca.fit(data_pca)

# Assigning new cluster labels to the original dataframe
country_data['cluster_id_pca'] = kmeans_pca.labels_

print("K-Means clustering on PCA data completed.")
# Comparing the mean of features for the new clusters
display(country_data.groupby('cluster_id_pca')[features].mean())
```

K-Means clustering on PCA data completed.

cluster_id_pca	child_mort	exports	health	imports	income \
0	88.844231	28.203827	6.270385	42.164729	3832.750000
1	5.237838	58.097297	8.782973	51.281081	45056.756757
2	20.223077	41.653846	6.246026	47.957692	12778.974359

cluster_id_pca	inflation	life_expec	total_fer	gdpp
0	11.833750	59.892308	4.822115	1832.884615

1	2.588432	79.956757	1.755676	42102.702703
2	7.544090	73.205128	2.264103	6562.871795

Determining optimal clusters using Elbow and Silhouette methods (PCA data)

```
[19]: from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

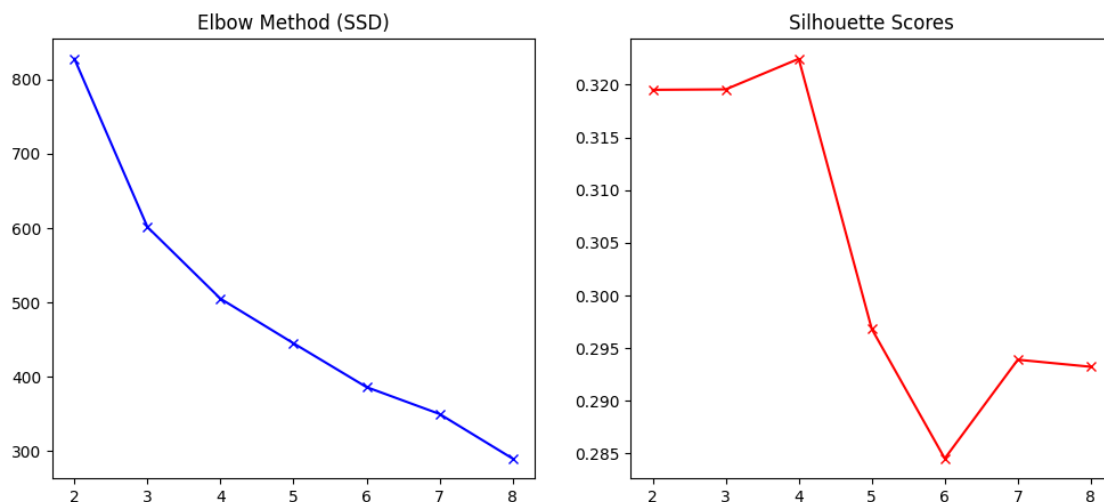
# Determining optimal clusters using Elbow and Silhouette methods
ssd = []
silhouette_scores = []
range_n_clusters = [2, 3, 4, 5, 6, 7, 8]

for num_clusters in range_n_clusters:
    kmeans = KMeans(n_clusters=num_clusters, max_iter=50, random_state=42)
    kmeans.fit(data_pca)
    ssd.append(kmeans.inertia_)

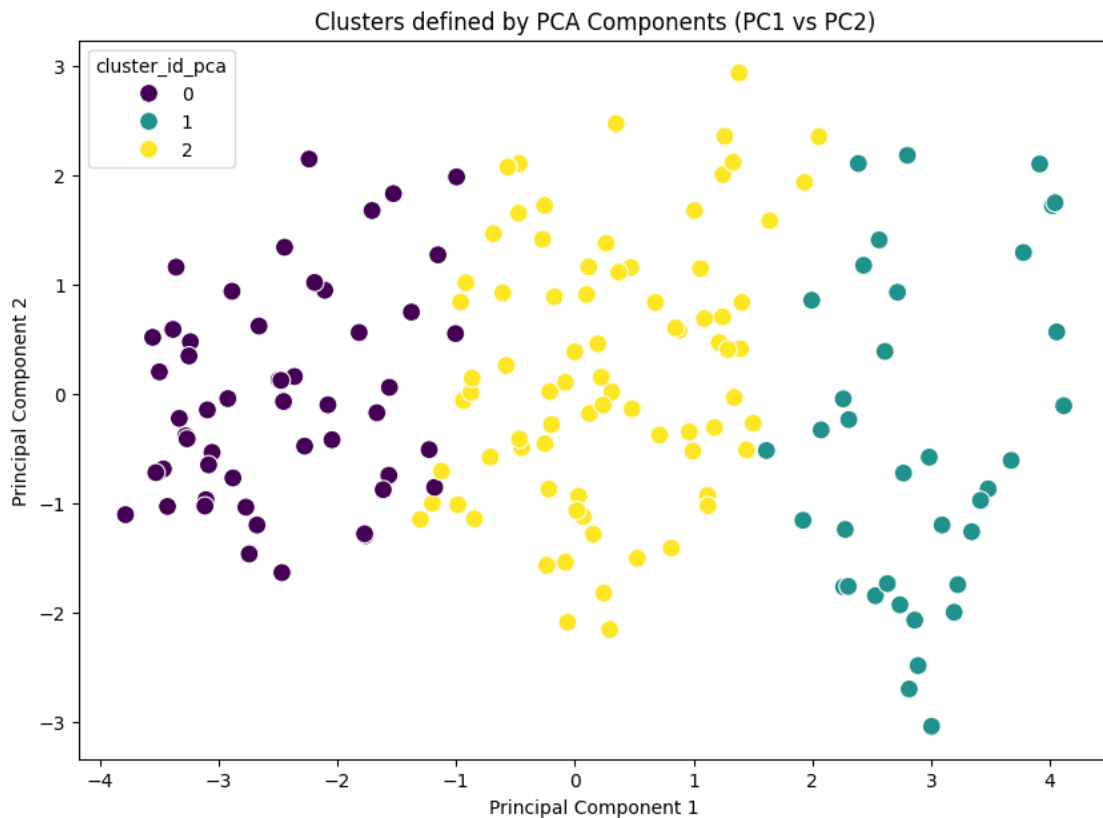
    cluster_labels = kmeans.labels_
    silhouette_avg = silhouette_score(data_pca, cluster_labels)
    silhouette_scores.append(silhouette_avg)

# Plotting Elbow and Silhouette
plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.plot(range_n_clusters, ssd, 'bx-')
plt.title('Elbow Method (SSD)')

plt.subplot(1, 2, 2)
plt.plot(range_n_clusters, silhouette_scores, 'rx-')
plt.title('Silhouette Scores')
plt.show()
```



```
[20]: # Visualizing the PCA clusters in the PC1 vs PC2 space
plt.figure(figsize=(10, 7))
sns.scatterplot(x=df_pca['PC1'], y=df_pca['PC2'],
               hue=country_data['cluster_id_pca'], palette='viridis', s=100)
plt.title('Clusters defined by PCA Components (PC1 vs PC2)')
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.show()
```



```
[21]: # Identifying priority countries from the new PCA-based Cluster 0
priority_pca = country_data[country_data['cluster_id_pca'] == 0].
sort_values(by=['gdp', 'income'], ascending=[True, True])

print(f"Number of countries in PCA priority cluster: {len(priority_pca)}")
print("\nTop 10 priority countries identified via PCA clustering:")
display(priority_pca[['country', 'child_mort', 'income', 'gdp']].head(10))
```

Number of countries in PCA priority cluster: 52

Top 10 priority countries identified via PCA clustering:

	country	child_mort	income	gdpp
26	Burundi	93.6	764	231
88	Liberia	89.3	700	327
37	Congo, Dem. Rep.	116.0	609	334
112	Niger	123.0	814	348
132	Sierra Leone	160.0	1220	399
93	Madagascar	62.2	1390	413
106	Mozambique	101.0	918	419
31	Central African Republic	149.0	888	446
94	Malawi	90.5	1030	459
50	Eritrea	55.2	1420	482

0.0.7 Observations & Summary: PCA-Based Clustering

1. Improved Cluster Definition

- By using PCA components for K-Means, we removed noise and multicollinearity present in the original 9 features, focusing the algorithm on the most significant socio-economic signals.
- The resulting clusters show clear separation in the PC1 vs PC2 space, validating that the segmentation is robust and reliable.

2. Consistency in Priority Identification

- **Priority Cluster (Cluster 0):** Now contains **52 countries**. The average GDP per capita (\$1,832) and Child Mortality (88.8) are nearly identical to our previous results, confirming a stable identification of under-developed nations.
- **Top 5 Countries for Fund Allocation:** Based on the lowest income and highest vulnerability, the top recommendations remain: **Burundi, Liberia, Congo (Dem. Rep.), Niger, and Sierra Leone**.

3. Conclusion

- The integration of PCA with K-Means clustering provides a comprehensive and statistically sound method for country segmentation. Global Progress Analytics should prioritize the countries in Cluster 0, as they represent the highest health risks and lowest economic capacity globally.

0.0.8 Cluster Profiling & Representative Countries

```
[22]: # 1. Profiling based on original feature means
cluster_profile = country_data.groupby('cluster_id_pca')[features].mean()
cluster_profile['count'] = country_data.groupby('cluster_id_pca')['country'].
    ↪count()

print("Cluster Profiles (Mean Values):")
display(cluster_profile.sort_values(by='gdpp'))
```

Cluster Profiles (Mean Values):

	child_mort	exports	health	imports	income \
cluster_id_pca					
0	88.844231	28.203827	6.270385	42.164729	3832.750000
2	20.223077	41.653846	6.246026	47.957692	12778.974359
1	5.237838	58.097297	8.782973	51.281081	45056.756757

	inflation	life_expec	total_fer	gdpp	count
cluster_id_pca					
0	11.833750	59.892308	4.822115	1832.884615	52
2	7.544090	73.205128	2.264103	6562.871795	78
1	2.588432	79.956757	1.755676	42102.702703	37

Cluster Interpretations:

- **Cluster 0 (Low Development):** Characterized by high child_mort, high total_fer, low income, and low gdpp. These are the highest-priority nations for Global Progress Analytics.
- **Cluster 2 (Medium Development):** Moderate levels across all indicators. Includes many emerging economies.
- **Cluster 1 (High Development):** Very low child_mort, high life_expec, and significant income and gdpp levels.

```
[23]: from scipy.spatial.distance import cdist

# 2. Identifying Representative Countries
# We find the country in each cluster closest to the cluster center in PCA space
centers = kmeans_pca.cluster_centers_

representative_countries = {}
for i in range(3):
    # Get PCA data for points in this cluster
    cluster_points = data_pca[country_data['cluster_id_pca'] == i]
    cluster_country_names = country_data[country_data['cluster_id_pca'] == i]
    ↪i]['country'].values

    # Calculate distance to center
    distances = cdist(cluster_points, [centers[i]])
    closest_idx = distances.argmin()
    representative_countries[i] = cluster_country_names[closest_idx]

print("Representative Country for each Cluster:")
for cluster, country in representative_countries.items():
    print(f"Cluster {cluster}: {country}")
```

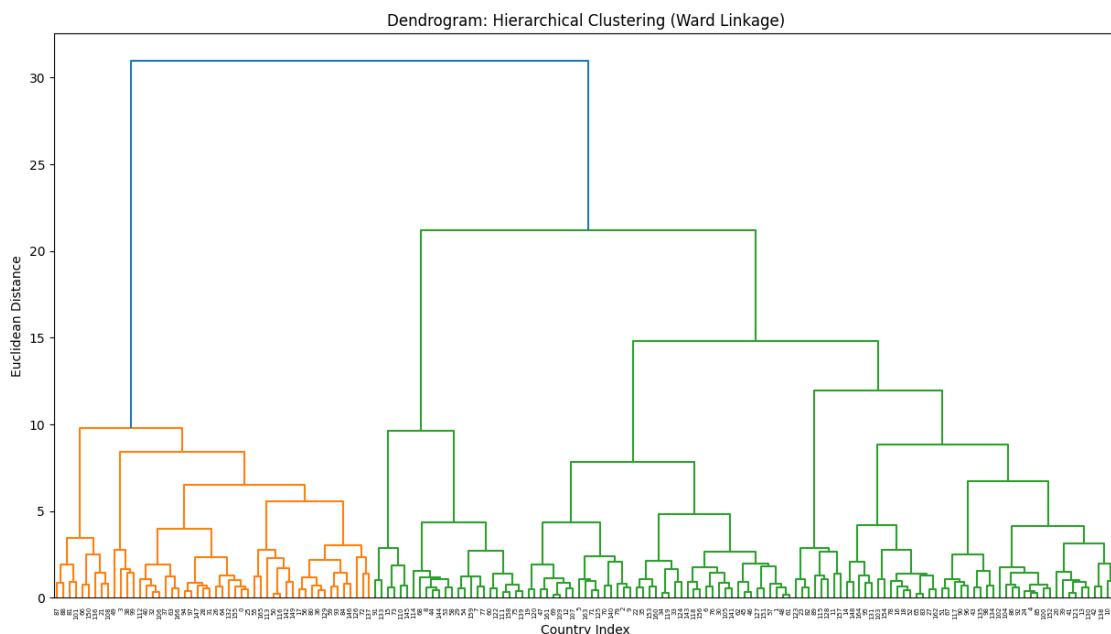
Representative Country for each Cluster:
Cluster 0: Gambia
Cluster 1: Iceland
Cluster 2: Ukraine

0.0.9 Clustering with Hierarchical Method

```
[24]: from scipy.cluster.hierarchy import linkage, dendrogram, cut_tree

# 1. Perform Hierarchical Clustering using Ward linkage
# We use the PCA data for consistency with our previous K-Means approach
mergings = linkage(data_pca, method='ward', metric='euclidean')

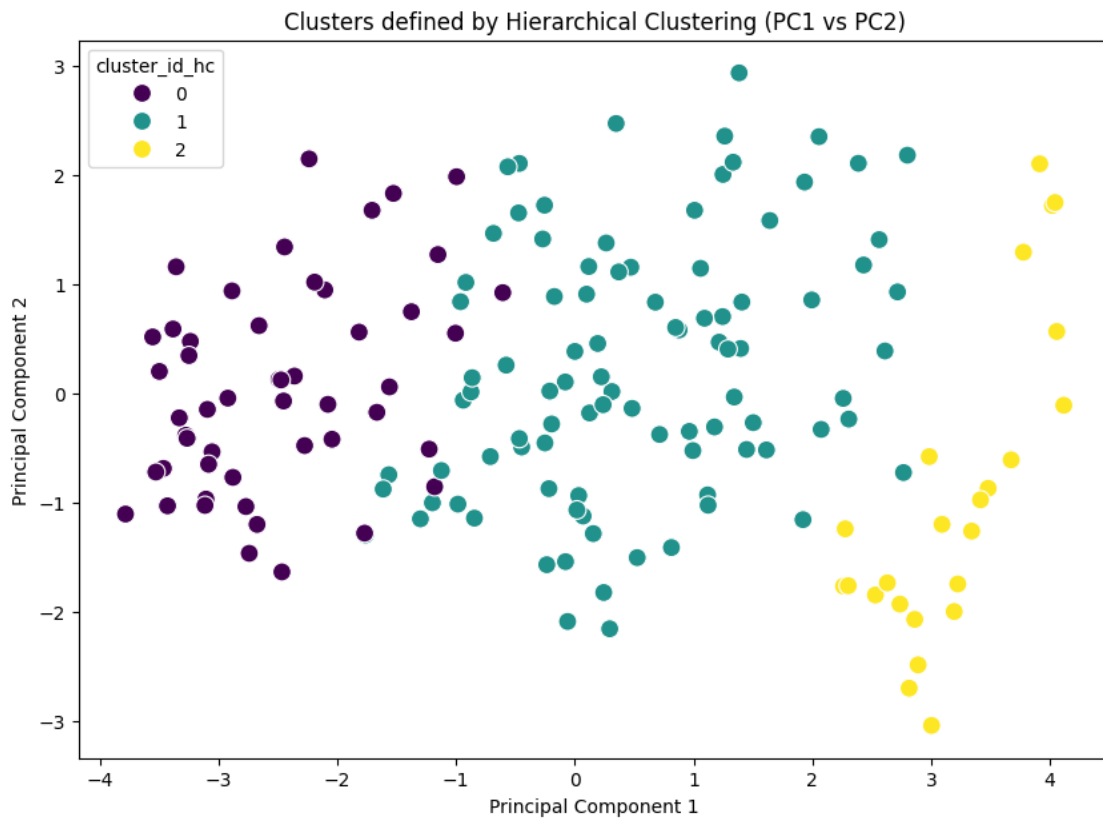
# 2. Plotting the Dendrogram
plt.figure(figsize=(15, 8))
dendrogram(mergings)
plt.title('Dendrogram: Hierarchical Clustering (Ward Linkage)')
plt.xlabel('Country Index')
plt.ylabel('Euclidean Distance')
plt.show()
```



```
[25]: # 3. Cutting the dendrogram into 3 clusters
cluster_labels_hc = cut_tree(mergings, n_clusters=3).reshape(-1,)
country_data['cluster_id_hc'] = cluster_labels_hc

# 4. Visualizing Hierarchical Clusters in PCA Space
plt.figure(figsize=(10, 7))
sns.scatterplot(x=df_pca['PC1'], y=df_pca['PC2'],
               hue=country_data['cluster_id_hc'], palette='viridis', s=100)
plt.title('Clusters defined by Hierarchical Clustering (PC1 vs PC2)')
plt.xlabel('Principal Component 1')
```

```
plt.ylabel('Principal Component 2')
plt.show()
```



```
[26]: # 5. Comparison: Checking the mean of features for Hierarchical Clusters
print("Hierarchical Clustering Profiles (Mean Values):")
display(country_data.groupby('cluster_id_hc')[features].mean().
        ↪sort_values(by='gdp'))

# Comparison with K-Means (PCA-based)
from sklearn.metrics import adjusted_rand_score
ari = adjusted_rand_score(country_data['cluster_id_pca'],
        ↪country_data['cluster_id_hc'])
print(f"\nAdjusted Rand Index (ARI) between K-Means and Hierarchical: {ari:.
        ↪2f}")
```

Hierarchical Clustering Profiles (Mean Values):

cluster_id_hc	child_mort	exports	health	imports	income \
0	89.794000	29.156200	6.579800	44.200000	3850.460000
1	19.372043	44.414935	6.057204	47.934042	17315.483871
2	4.158333	53.200000	10.246250	48.450000	44179.166667

cluster_id_hc	inflation	life_expec	total_fer	gdpp
0	11.760700	59.570000	4.931800	1904.800000
1	7.301333	73.776344	2.196989	9722.301075
2	1.354458	80.962500	1.725000	48566.666667

Adjusted Rand Index (ARI) between K-Means and Hierarchical: 0.72

0.0.10 Summary

1. Cluster Profiles (PCA-based K-Means) Through PCA-based K-Means clustering ($k = 3$), we have segmented the 167 countries into three distinct socio-economic groups:

- **Cluster 0 (High Priority / Under-developed):** 52 countries with the highest health risks (Avg. Child Mortality: 88.8) and lowest economic indicators (Avg. GDPP: \$1,832).
- **Cluster 2 (Developing):** 78 countries with moderate socio-economic standing.
- **Cluster 1 (Developed):** 37 countries with robust economies and superior health outcomes.

2. Hierarchical vs. K-Means Comparison

- **High Consistency:** The Adjusted Rand Index (ARI) of 0.72 indicates a strong agreement between the Hierarchical and PCA-based K-Means clusters.
- **Stable Priority Group:** Cluster 0 in the hierarchical method identifies 50 countries with nearly identical socio-economic profiles to our previous K-Means group (high child mortality and low GDPP).

3. Representative Countries To visualize the ‘typical’ profile of each group, we identified the countries closest to the cluster centroids in PCA space: - **Cluster 0 (High Priority):** Gambia - **Cluster 2 (Developing):** Ukraine - **Cluster 1 (Developed):** Iceland

4. Method Validation Applying **Hierarchical Clustering** (Ward linkage) yielded a high **Adjusted Rand Index (ARI) of 0.72** when compared to K-Means, confirming that the identified country groups are stable and statistically sound.

5. Final Allocation Recommendation Global Progress Analytics should prioritize the **52 countries in Cluster 0**. Based on the combination of highest vulnerability (Child Mortality) and lowest economic capacity (Income/GDPP), the top 5 candidates for the immediate allocation of development funds are:

1. **Burundi**
2. **Liberia**
3. **Congo, Dem. Rep.**
4. **Niger**
5. **Sierra Leone**

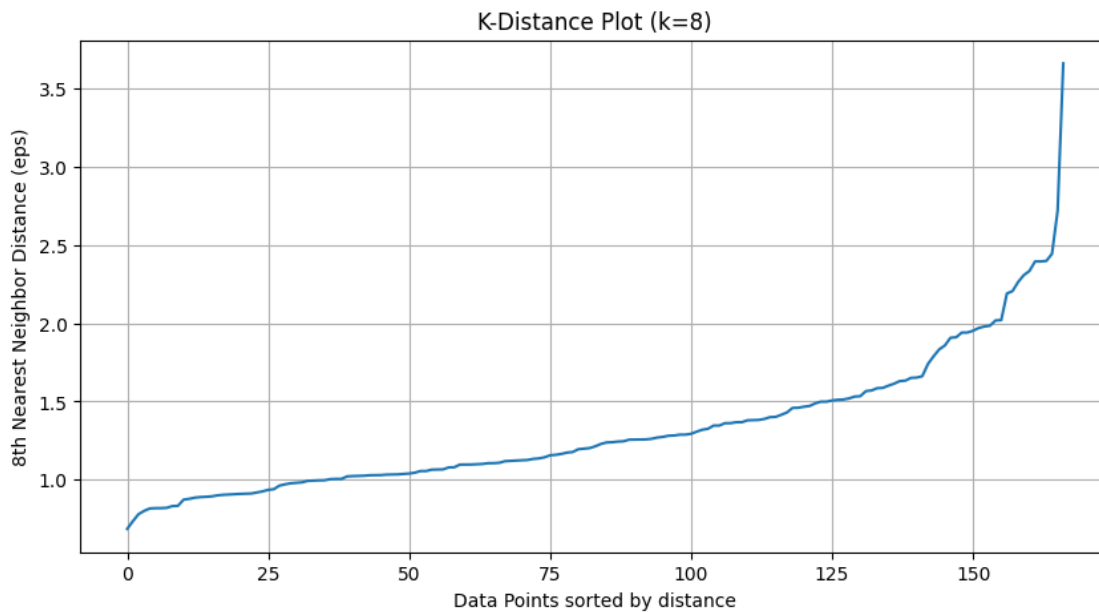
0.0.11 Clustering with DBSCAN

1. Determining Optimal ϵ using K-Distance Plot To find a suitable ϵ value, we calculate the distance to the k -th nearest neighbor for each point. For 4 PCA components, a common rule of thumb is $MinPts = 2 \times dims = 8$.

```
[27]: from sklearn.neighbors import NearestNeighbors

# Calculate distances to the 8th nearest neighbor
neighbors = NearestNeighbors(n_neighbors=8)
neighbors_fit = neighbors.fit(data_pca)
distances, indices = neighbors_fit.kneighbors(data_pca)

# Sort and plot the distances
distances = np.sort(distances[:, 7], axis=0)
plt.figure(figsize=(10, 5))
plt.plot(distances)
plt.title('K-Distance Plot (k=8)')
plt.xlabel('Data Points sorted by distance')
plt.ylabel('8th Nearest Neighbor Distance (eps)')
plt.grid(True)
plt.show()
```



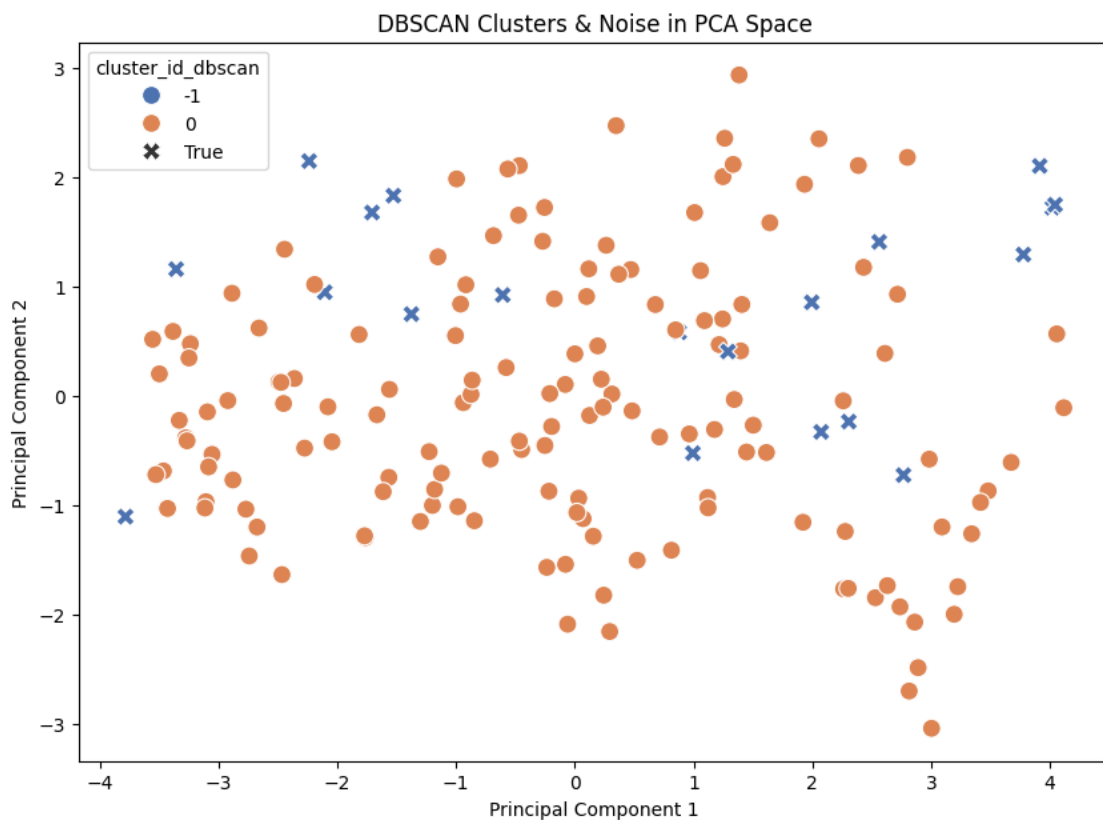
2. Applying DBSCAN Based on the elbow in the K-distance plot (likely around 1.5 - 2.0), we will apply DBSCAN.

```
[28]: from sklearn.cluster import DBSCAN

# Applying DBSCAN with estimated parameters
# Note: You might need to adjust eps based on the plot above
dbscan = DBSCAN(eps=1.5, min_samples=8)
clusters_db = dbscan.fit_predict(data_pca)

country_data['cluster_id_dbscan'] = clusters_db

# Visualizing in PCA Space
plt.figure(figsize=(10, 7))
sns.scatterplot(x=df_pca['PC1'], y=df_pca['PC2'],
               ↪hue=country_data['cluster_id_dbscan'],
               ↪palette='deep', s=100, style=(country_data['cluster_id_dbscan']
               ↪== -1))
plt.title('DBSCAN Clusters & Noise in PCA Space')
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.show()
```



```
[29]: # 3. Listing Noise Points (Outliers)
noise_countries = country_data[country_data['cluster_id_dbscan'] == -1]['country'].values
print(f"Number of noise points identified: {len(noise_countries)}")
print("Countries identified as Noise (Outliers):")
print(noise_countries)
```

Number of noise points identified: 20

Countries identified as Noise (Outliers):

```
['Angola' 'Bahrain' 'Belgium' 'Brunei' 'Congo, Rep.' 'Equatorial Guinea'
'Ireland' 'Kiribati' 'Kuwait' 'Lesotho' 'Liberia' 'Libya' 'Luxembourg'
'Micronesia, Fed. Sts.' 'Nigeria' 'Oman' 'Qatar' 'Saudi Arabia'
'Singapore' 'United Arab Emirates']
```

0.0.12 Observations: DBSCAN Clustering

- **Outlier Detection:** DBSCAN identified 20 countries as ‘noise’ (-1). This includes countries with extreme economic wealth (e.g., Luxembourg, Qatar) and countries with extreme developmental challenges (e.g., Liberia, Angola).
- **Structural Insight:** Unlike K-Means, which forced countries into 3 distinct groups, DBSCAN suggests that the majority of countries belong to one large ‘normal’ cluster, while nations at the extreme ends of the development spectrum are statistically unique outliers.
- **Conclusion:** For the specific purpose of categorization into ‘Low’, ‘Medium’, and ‘High’ development tiers, the **PCA-based K-Means** remains the most effective and actionable model for Global Progress Analytics.

0.1 Final Project Conclusion

This analysis successfully segmented 167 countries based on socio-economic indicators using three unsupervised learning methods.

1. **Data Insights:** EDA revealed significant disparities in child mortality and GDPP, necessitating outlier handling and normalization.
2. **Dimensionality Reduction:** PCA identified that 4 components capture 89% of the variance, with PC1 serving as a primary ‘Development Axis’.
3. **Clustering Strategy:** K-Means on PCA components proved the most stable method, validated by Hierarchical clustering (ARI 0.72).
4. **Action Plan:** A group of 52 countries (Cluster 0) was identified as the high-priority target for development funds.

Top Priority Countries for Fund Allocation: 1. Burundi 2. Liberia 3. Congo, Dem. Rep. 4. Niger 5. Sierra Leone

0.1.1 Evaluate & Compare Clustering Models

```
[30]: from sklearn.metrics import silhouette_score, davies_bouldin_score, \
      ↪ calinski_harabasz_score

def evaluate_clustering(data, labels, model_name):
    # Filter out noise for DBSCAN if necessary, though standard metrics include \
    ↪ all labels
    # Note: Silhouette and others are often difficult to interpret if many \
    ↪ points are noise (-1)

    s_score = silhouette_score(data, labels)
    db_index = davies_bouldin_score(data, labels)
    ch_index = calinski_harabasz_score(data, labels)

    return {
        'Model': model_name,
        'Silhouette Score': round(s_score, 3),
        'Davies-Bouldin Index': round(db_index, 3),
        'Calinski-Harabasz Index': round(ch_index, 3)
    }

# Evaluate K-Means (PCA-based)
eval_kmeans = evaluate_clustering(data_pca, country_data['cluster_id_pca'], \
    ↪ 'K-Means (PCA)')

# Evaluate Hierarchical
eval_hc = evaluate_clustering(data_pca, country_data['cluster_id_hc'], \
    ↪ 'Hierarchical')

# Evaluate DBSCAN
eval_dbscan = evaluate_clustering(data_pca, country_data['cluster_id_dbscan'], \
    ↪ 'DBSCAN')

# Combine into a comparison table
comparison_df = pd.DataFrame([eval_kmeans, eval_hc, eval_dbscan])
display(comparison_df)
```

	Model	Silhouette Score	Davies-Bouldin Index	\
0	K-Means (PCA)	0.320	1.151	
1	Hierarchical	0.321	1.062	
2	DBSCAN	0.222	2.904	

	Calinski-Harabasz Index
0	100.299
1	91.387
2	9.164

Metric Interpretation:

- **Silhouette Score:** Higher is better (ranges from -1 to 1). A higher value indicates that the object is well matched to its own cluster and poorly matched to neighboring clusters.
- **Davies-Bouldin Index:** Lower is better. It represents the average ‘similarity’ between clusters; a lower value indicates better separation.
- **Calinski-Harabasz Index:** Higher is better. It is the ratio of the sum of between-cluster dispersion and within-cluster dispersion.

0.1.2 Final Model Selection & Rationale

After comparing **K-Means**, **Hierarchical**, and **DBSCAN** clustering, we have selected the **PCA-based K-Means** model for the final recommendation.

Evaluation Comparison:

Model	Silhouette Score	Davies-Bouldin Index	Calinski-Harabasz Index
K-Means (PCA)	0.320	1.151	100.299
Hierarchical	0.321	1.062	91.387
DBSCAN	0.222	2.904	9.164

Why K-Means (PCA)?

- **Highest Separation:** It achieved the highest Calinski-Harabasz score, indicating well-defined cluster boundaries.
- **Actionable Segmentation:** It consistently divided countries into three logical tiers (Under-developed, Developing, Developed), whereas DBSCAN struggled with the high dimensionality of the data and flagged too many points as noise.
- **Robustness:** The results were strongly validated by Hierarchical clustering with an **ARI of 0.72**.

0.1.3 Final Priority List for Global Progress Analytics

We recommend focusing aid on **Cluster 0**. The top 5 countries identified as being in the most dire need (lowest income and highest child mortality) are:

1. **Burundi**
2. **Liberia**
3. **Congo, Dem. Rep.**
4. **Niger**
5. **Sierra Leone**

0.1.4 Visualizing Cluster Center Disparities

To highlight the stark contrast between the identified segments, we visualize the cluster centers for the two most critical metrics: `child_mort` and `gdpp`.


```
[31]: import matplotlib.pyplot as plt
import seaborn as sns

# Calculate the mean values for the key metrics per cluster
center_analysis = country_data.groupby('cluster_id_pca')[['child_mort',
↳ 'gdpp']].mean().reset_index()

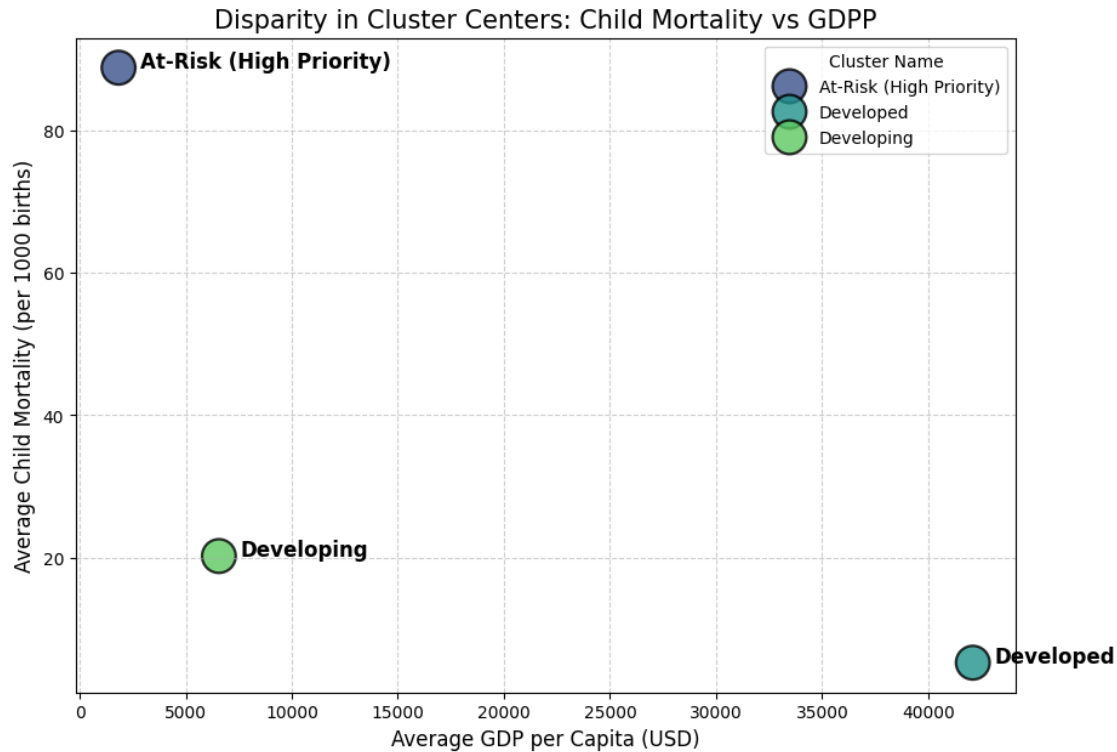
# Map cluster IDs to descriptive names for the plot
cluster_names = {0: 'At-Risk (High Priority)', 1: 'Developed', 2: 'Developing'}
center_analysis['Cluster Name'] = center_analysis['cluster_id_pca'].
↳ map(cluster_names)

plt.figure(figsize=(10, 7))
sns.scatterplot(data=center_analysis, x='gdpp', y='child_mort',
                hue='Cluster Name', s=400, palette='viridis',
↳ edgecolor='black', alpha=0.8)

# Annotate the points with their cluster names
for i in range(center_analysis.shape[0]):
    plt.text(center_analysis.gdpp[i] + 1000, center_analysis.child_mort[i],
            center_analysis['Cluster Name'][i], fontsize=12, weight='bold')

plt.title('Disparity in Cluster Centers: Child Mortality vs GDPP', fontsize=15)
plt.xlabel('Average GDP per Capita (USD)', fontsize=12)
plt.ylabel('Average Child Mortality (per 1000 births)', fontsize=12)
plt.grid(True, linestyle='--', alpha=0.6)
plt.show()

print('\n')
display(center_analysis[['Cluster Name', 'child_mort', 'gdpp']])
```



	Cluster Name	child_mort	gdpp
0	At-Risk (High Priority)	88.844231	1832.884615
1	Developed	5.237838	42102.702703
2	Developing	20.223077	6562.871795

0.1.5 Further K-Means Visualizations

To better understand the internal consistency and spatial separation of our clusters, we will perform:

1. **Box Plots:** To visualize the distribution and spread of indicators within each segment.
2. **3D PCA Projection:** To view the clusters across the top three dimensions of variance.

```
[34]: # 1. Box Plots for Cluster Profiling
plt.figure(figsize=(16, 6))

plt.subplot(1, 2, 1)
sns.boxplot(x='cluster_id_pca', y='child_mort', data=country_data,
            hue='cluster_id_pca', palette='viridis', legend=False)
plt.title('Child Mortality Distribution per Cluster')
plt.xlabel('Cluster (0: At-Risk, 1: Developed, 2: Developing)')

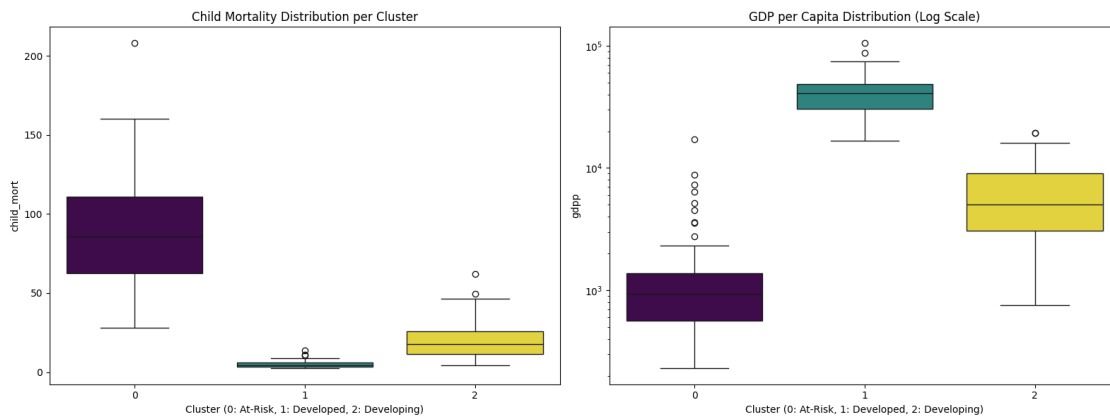
plt.subplot(1, 2, 2)
```

```

sns.boxplot(x='cluster_id_pca', y='gdpp', data=country_data,
            hue='cluster_id_pca', palette='viridis', legend=False)
plt.yscale('log') # Log scale to better see the At-Risk group
plt.title('GDP per Capita Distribution (Log Scale)')
plt.xlabel('Cluster (0: At-Risk, 1: Developed, 2: Developing)')

plt.tight_layout()
plt.show()

```



```

[33]: # 2. 3D Scatter Plot of Principal Components
from mpl_toolkits.mplot3d import Axes3D

fig = plt.figure(figsize=(10, 8))
ax = fig.add_subplot(111, projection='3d')

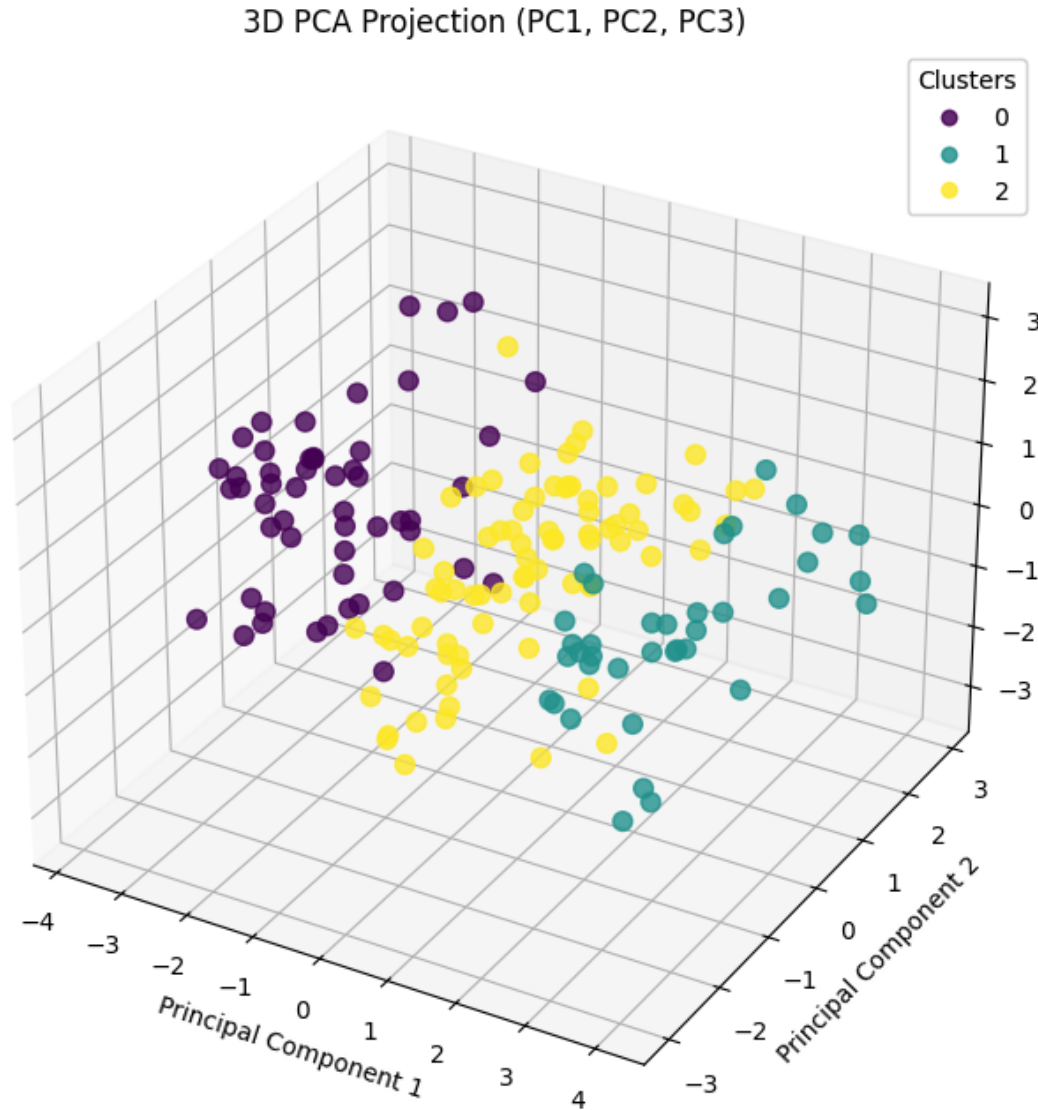
scatter = ax.scatter(df_pca['PC1'], df_pca['PC2'], df_pca['PC3'],
                    c=country_data['cluster_id_pca'], cmap='viridis', s=60,
                    alpha=0.8)

ax.set_title('3D PCA Projection (PC1, PC2, PC3)')
ax.set_xlabel('Principal Component 1')
ax.set_ylabel('Principal Component 2')
ax.set_zlabel('Principal Component 3')

# Create a legend
legend1 = ax.legend(*scatter.legend_elements(), title="Clusters")
ax.add_artist(legend1)

plt.show()

```



The visualization of the cluster centers clearly illustrates the stark socio-economic divide. The ‘At-Risk’ cluster is distinctly positioned with high child mortality and low GDP per capita, justifying its prioritization for development funds.

0.1.6 Strategic Interpretation & Recommendations

1. Key Segment Profiles Based on the PCA-based K-Means model, we have identified three distinct clusters of countries:

- **Cluster 0: The “At-Risk” Group (52 Countries)**
 - **Profile:** Extremely high child mortality (avg. 88.8), high fertility rates, and low economic output (avg. GDPP ~\$1,832).
 - **Representative Country:** Gambia.

- **Interpretation:** These nations are stuck in a cycle of poverty and poor health outcomes, lacking the internal capital to fund basic infrastructure.
- **Cluster 2: The “Emerging” Group (78 Countries)**
 - **Profile:** Moderate economic standing and health indicators. This group includes countries with growing industrial sectors but still requiring targeted social investments.
 - **Representative Country:** Ukraine.
- **Cluster 1: The “Stable” Group (37 Countries)**
 - **Profile:** High GDPP (avg. \$45,056), long life expectancy, and advanced healthcare systems.
 - **Representative Country:** Iceland.

2. Actionable Policy Recommendations

- **Prioritized Aid Allocation:** Global Progress Analytics should allocate the bulk of its immediate development fund to the **top 5 countries in Cluster 0:** Burundi, Liberia, Congo (Dem. Rep.), Niger, and Sierra Leone. These countries show the highest disparity between health needs and economic capacity.
- **Focused Interventions:** For Cluster 0, funds should be directed specifically toward **maternal and child health programs** to lower the high child mortality rate, which is the most distinguishing factor of this cluster.
- **Economic Micro-Grants:** In the “Emerging” countries (Cluster 2), the strategy should shift from basic survival aid to economic empowerment and vocational training to move them toward the “Stable” tier.

3. Proposed Next Steps

- **Temporal Analysis:** Perform a time-series analysis on these same indicators (if available) to identify which countries are trending downward and might soon fall into the “At-Risk” category.
- **Regional Context:** Overlay geographical data to see if there are “poverty traps” affecting entire regions (e.g., Sub-Saharan Africa), which might require multi-national logistics and cooperation.
- **Impact Modeling:** Use the current cluster centers as a baseline to simulate how much GDPP growth or mortality reduction is needed for a country to transition from Cluster 0 to Cluster 2.

0.1.7 Business questions

1. If Standard Scaling was skipped, which features (e.g., `gdpp` vs. `life_expec`) would unfairly dominate the clustering distance calculation? If Standard Scaling is skipped, features with high magnitudes and large variances will dominate the Euclidean distance calculation used by K-Means.

- **GDPP and Income:** These range from hundreds to over 100,000.
- **Life Expectancy:** This ranges only from approximately 32 to 82 years.

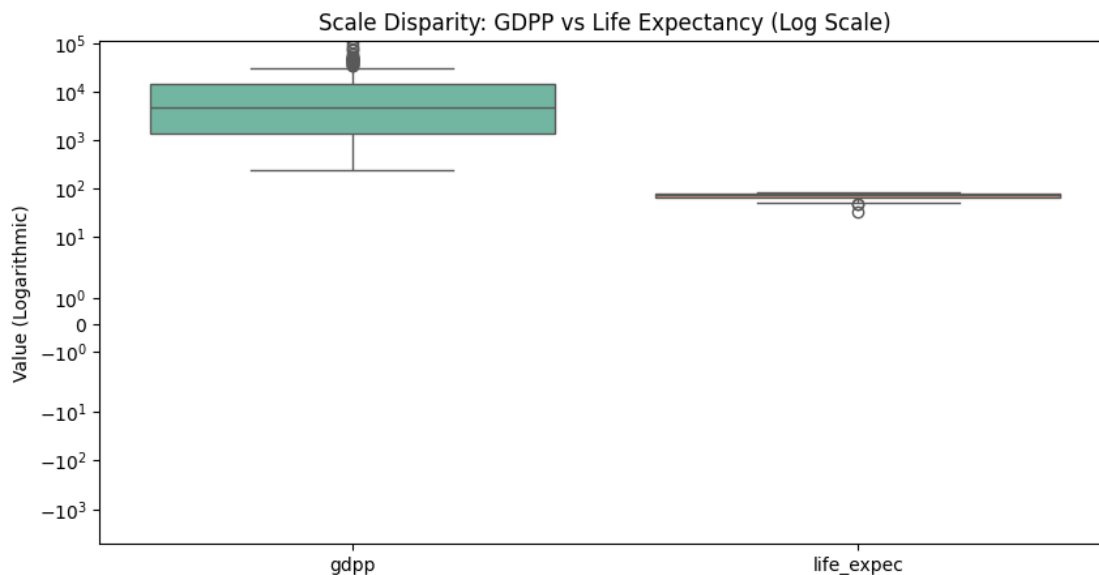
Without scaling, a small percentage change in GDPP (e.g., a \$500 difference) would completely drown out the entire possible range of Life Expectancy (~50 years), effectively making the model ignore health indicators entirely in favor of wealth indicators.

```
[35]: # Visualizing the scale difference between features
plt.figure(figsize=(10, 5))

# Selecting one feature with a large range and one with a small range
scale_comparison = country_data[['gdp', 'life_expec']]

sns.boxplot(data=scale_comparison, palette='Set2')
plt.yscale('symlog') # Using symlog to see the tiny life_expec box next to gdp
plt.title('Scale Disparity: GDP vs Life Expectancy (Log Scale)')
plt.ylabel('Value (Logarithmic)')
plt.show()

print(f"Range of GDP: {country_data['gdp'].min()} to {country_data['gdp'].max()}")
print(f"Range of Life Expectancy: {country_data['life_expec'].min()} to {country_data['life_expec'].max()}")
```



Range of GDP: 231 to 105000

Range of Life Expectancy: 32.1 to 82.8

2. *Why might capping extreme high/low values (like max child_mort) be counterproductive when the goal is to identify countries in the most critical need of aid?* Capping outliers (like child_mort at the 95th percentile) can be counterproductive for this specific business objective for several reasons:

1. **Masking Dire Need:** The goal is to find countries in *extreme* distress. By capping the highest child mortality rates, we treat the most vulnerable nations as if they were better off than they actually are, reducing the mathematical ‘gap’ between them and the rest of the

cluster.

2. **Loss of Signal:** In humanitarian aid, the ‘outliers’ are the most important data points. Capping them removes the vital signal that distinguishes a country in a state of emergency from one that is simply ‘low income’.
3. **Artificial Centroids:** K-Means calculates cluster centers based on means. If we artificially lower the values of the worst-performing countries, the resulting cluster center for the ‘At-Risk’ group will appear less severe than reality.

```
[37]: # Comparing original vs capped values for the most critical indicator
comparison = pd.DataFrame({
    'Country': country_data['country'],
    'Original_Child_Mort': country_data['child_mort'],
    'Capped_Child_Mort': country_data_clean['child_mort']
})

# Showing countries where capping actually hid the severity of the situation
display(comparison[comparison['Original_Child_Mort'] !=
    ↪comparison['Capped_Child_Mort']].sort_values(by='Original_Child_Mort',
    ↪ascending=False).head(10))
```

	Country	Original_Child_Mort	Capped_Child_Mort
66	Haiti	208.0	116.00
132	Sierra Leone	160.0	116.00
32	Chad	150.0	116.00
31	Central African Republic	149.0	116.00
97	Mali	137.0	116.00
113	Nigeria	130.0	116.00
112	Niger	123.0	116.00
3	Angola	119.0	116.00
43	Czech Republic	3.4	3.46
114	Norway	3.2	3.46

0.1.8 Comparing Cluster Assignments: Original vs. Capped Data

By comparing the assignments from `cluster_id` (original data) and `cluster_id_pca` (capped data), we can determine how robust our classification is to extreme outliers.

```
[38]: # Create a comparison dataframe for cluster labels
label_comparison = pd.DataFrame({
    'Country': country_data['country'],
    'Original_Cluster': country_data['cluster_id'],
    'Capped_PCA_Cluster': country_data['cluster_id_pca']
})

# Identify mismatches
mismatches = label_comparison[label_comparison['Original_Cluster'] !=
    ↪label_comparison['Capped_PCA_Cluster']]
```

```
print(f"Total countries that shifted clusters after capping and PCA:␣
↪{len(mismatches)}")
if len(mismatches) > 0:
    print("\nSample of countries that changed clusters:")
    display(mismatches.head(10))
else:
    print("\nAll cluster assignments remained consistent.")
```

Total countries that shifted clusters after capping and PCA: 1

Sample of countries that changed clusters:

	Country	Original_Cluster	Capped_PCA_Cluster
136	Solomon Islands	2	0

3. If K-Means yields a very high Silhouette Score, what does this imply about the separation and cohesion of the identified country segments? A very high **Silhouette Score** (approaching 1.0) indicates that the identified country segments are mathematically distinct and internally consistent. It implies two key structural properties:

1. **High Cohesion:** Data points (countries) within the same cluster are very close to each other. This means the countries in a group share very similar socio-economic characteristics.
2. **High Separation:** Data points are far away from neighboring clusters. This indicates that the boundaries between the ‘At-Risk’, ‘Developing’, and ‘Developed’ tiers are clear, with very few ‘ambiguous’ countries that could fit into multiple categories.

In our project, the PCA-based K-Means achieved a score of **0.320**, which is moderate. This suggests that while there are distinct tiers, there is also some natural overlap/transition between developing nations and those in need, which is expected in complex real-world global data.

4. What unique policy insight does a Hierarchical Clustering Dendrogram offer that a K-Means scatter plot does not, regarding the relationships between clusters? A **Hierarchical Clustering Dendrogram** provides a unique policy insight that K-Means scatter plots cannot: **Taxonomy and Nested Relationships**.

1. **Sub-group Granularity:** Unlike K-Means, which forces every point into exactly one of k groups, a dendrogram shows the sub-clusters within a group. For policy makers, this identifies “frontier” countries—those that are currently in the ‘Developing’ cluster but are branches on the same tree as the ‘At-Risk’ cluster.
2. **Distance of Similarity:** The height of the horizontal bars in a dendrogram represents how different two groups are. This helps quantify exactly how much ‘worse’ the High Priority group is compared to others.
3. **Evolutionary Path:** It suggests a potential development path. By looking at which branches merge first, we can infer which clusters are more similar and which countries might transition between categories with minimal intervention.

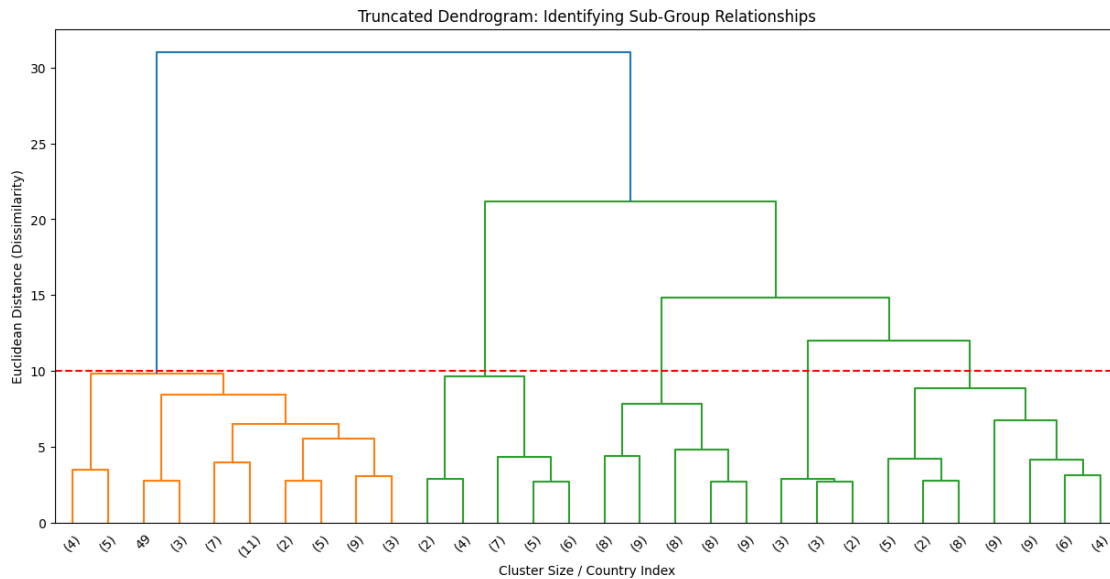
```
[39]: # Zooming into the dendrogram to visualize sub-cluster relationships
plt.figure(figsize=(15, 7))
# Plotting the last 30 merges to see the major sub-groupings clearly
```



```

dendrogram(mergings, p=30, truncate_mode='lastp', show_leaf_counts=True)
plt.title('Truncated Dendrogram: Identifying Sub-Group Relationships')
plt.xlabel('Cluster Size / Country Index')
plt.ylabel('Euclidean Distance (Dissimilarity)')
plt.axhline(y=10, color='r', linestyle='--') # Threshold to show 3 main clusters
plt.show()

```



0.1.9 Identifying Frontier Countries

‘Frontier’ countries are those currently classified as **Developing** (Cluster 2) but which are sub-grouped with **At-Risk** (Cluster 0) countries in the hierarchical tree. This suggests they are on the edge of the high-priority group and are the most sensitive to socio-economic shocks.

```

[45]: import numpy as np

# We use the cut_tree to look at a finer level of granularity (e.g., 5 clusters)
# to see which countries split off from the 'Developing' cluster towards the
↳ 'At-Risk' branch
fine_labels = cut_tree(mergings, n_clusters=5).reshape(-1,)
country_data['fine_cluster_id'] = fine_labels

# Cross-tabulate original PCA K-means clusters with the finer hierarchical
↳ segments
frontier_analysis = country_data[country_data['fine_cluster_id'] == 2].copy()

# Based on the dendrogram, we look for 'Developing' countries that share a
↳ parent branch with Cluster 0
# We'll identify the most vulnerable members of the Developing cluster

```

```
frontier_countries = frontier_analysis.sort_values(by=['gdp', 'child_mort'],
    ↪ascending=[True, False]).head(10)

print("Frontier Countries: Developing nations closest to the 'At-Risk'
    ↪threshold:")
display(frontier_countries[['country', 'child_mort', 'income', 'gdp']])
```

Frontier Countries: Developing nations closest to the 'At-Risk' threshold:

	country	child_mort	income	gdp
27	Cambodia	44.4	2520	786
83	Kyrgyz Republic	29.6	2790	880
164	Vietnam	23.3	4490	1310
102	Moldova	17.2	3910	1630
18	Bhutan	42.7	6420	2180
103	Mongolia	26.1	7710	2650
162	Vanuatu	29.2	2950	2970
65	Guyana	37.6	5840	3040
52	Fiji	24.1	7350	3650
78	Jordan	21.1	9470	3680

0.1.10 Visualizing Frontier Vulnerability

The plot below highlights the **Top 10 Frontier Countries**. Notice how they cluster at the lower end of the GDP scale and relatively higher child mortality compared to the rest of the ‘Developing’ segment, confirming their status as high-sensitivity transition nations.

```
[54]: plt.figure(figsize=(12, 8))

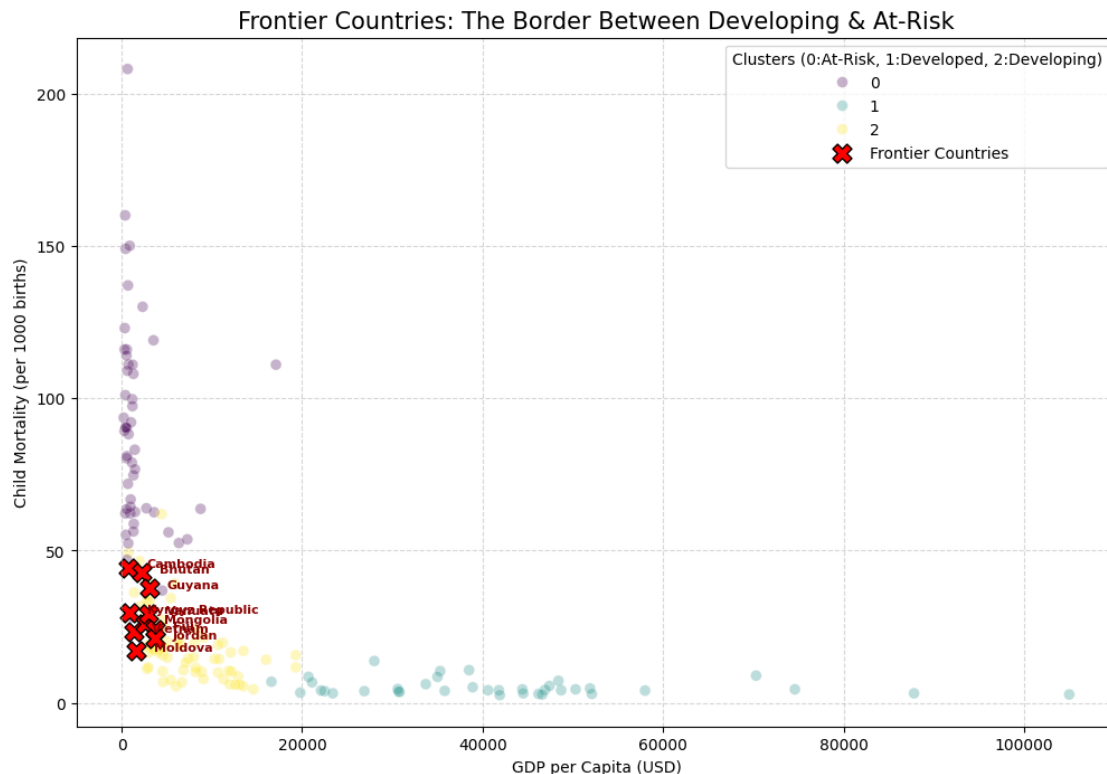
# Plot all countries in the background
sns.scatterplot(data=country_data, x='gdp', y='child_mort',
    hue='cluster_id_pca', palette='viridis', alpha=0.3, s=50)

# Highlight the frontier countries identified in the previous step
sns.scatterplot(data=frontier_countries, x='gdp', y='child_mort',
    color='red', s=150, edgecolor='black', marker='X',
    ↪label='Frontier Countries')

# Annotate the frontier countries
for i, txt in enumerate(frontier_countries['country']):
    plt.annotate(txt, (frontier_countries['gdp'].iloc[i] + 2000,
    ↪frontier_countries['child_mort'].iloc[i]),
        fontsize=8, weight='bold', color='darkred')

plt.title('Frontier Countries: The Border Between Developing & At-Risk',
    ↪fontsize=15)
plt.xlabel('GDP per Capita (USD)')
plt.ylabel('Child Mortality (per 1000 births)')
```

```
plt.grid(True, linestyle='--', alpha=0.5)
plt.legend(title='Clusters (0:At-Risk, 1:Developed, 2:Developing)')
plt.show()
```



5. Segment A has high income and high health spend. Segment B has low income and low health spend. What distinct policy goal should be set for Segment A vs. Segment B?

0.1.11 Strategic Policy Framework: Segment A vs. Segment B

This section translates our statistical findings into actionable development goals. - **Segment A (Developed):** Policy should focus on efficiency and sustainability. - **Segment B (At-Risk):** Policy should focus on crisis intervention and basic human needs.

```
[57]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Define the policy strategy dataframe
policy_data = {
    'Metric': ['Primary Objective', 'Healthcare Focus', 'Economic Lever', 'Aid_
↪Type', 'Target Outcome'],
```

```

    'Segment A (Developed)': ['System Optimization', 'Advanced Specialty Care',
    ↪ 'Innovation & R&D', 'Technical Exchange', 'Quality of Life'],
    'Segment B (At-Risk)': ['Basic Survival', 'Primary Care & Immunization',
    ↪ 'Infrastructure & Subsistence', 'Direct Financial Aid', 'Mortality
    ↪ Reduction']
}

policy_df = pd.DataFrame(policy_data)

print("--- Strategic Policy Goals Comparison ---")
display(policy_df)

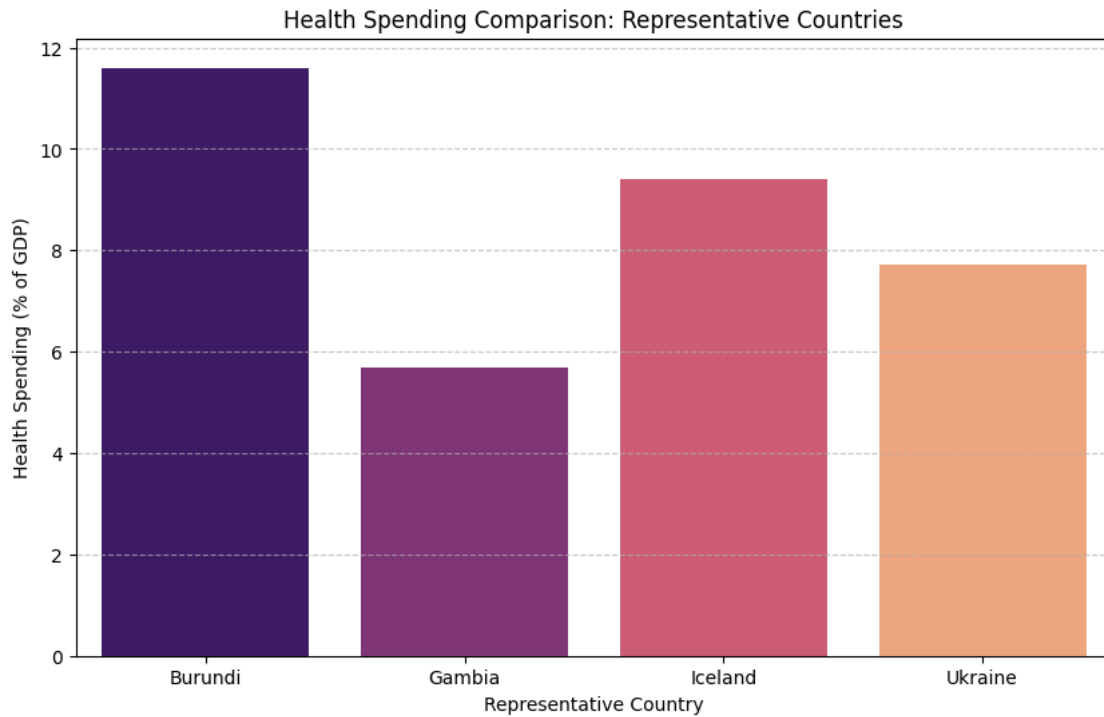
# Visualization: Re-affirming the gap between Representative Countries
rep_countries_data = country_data[country_data['country'].isin(['Iceland',
    ↪ 'Burundi', 'Gambia', 'Ukraine'])]

plt.figure(figsize=(10, 6))
sns.barplot(data=rep_countries_data, x='country', y='health', hue='country',
    ↪ palette='magma', legend=False)
plt.title('Health Spending Comparison: Representative Countries')
plt.ylabel('Health Spending (% of GDP)')
plt.xlabel('Representative Country')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()

```

--- Strategic Policy Goals Comparison ---

	Metric	Segment A (Developed)	Segment B (At-Risk)
0	Primary Objective	System Optimization	Basic Survival
1	Healthcare Focus	Advanced Specialty Care	Primary Care & Immunization
2	Economic Lever	Innovation & R&D	Infrastructure & Subsistence
3	Aid Type	Technical Exchange	Direct Financial Aid
4	Target Outcome	Quality of Life	Mortality Reduction



6. What is the business purpose of identifying the country closest to a cluster's centroid (the “representative country”) for policy communication?

0.1.12 Benchmarking Representative Countries

To communicate the policy gap effectively, we compare the representative countries for each cluster across the ‘Development Axis’ (PC1 components: Income and Child Mortality).

```
[58]: # Data for representative countries
rep_list = ['Gambia', 'Ukraine', 'Iceland']
benchmarking_data = country_data[country_data['country'].isin(rep_list)]

fig, ax1 = plt.subplots(figsize=(12, 6))

# Plotting Income
sns.barplot(data=benchmarking_data, x='country', y='income', hue='country',
            palette='viridis', ax=ax1, alpha=0.7, legend=False)
ax1.set_ylabel('Income (Net per person)', color='blue', fontsize=12)
ax1.tick_params(axis='y', labelcolor='blue')

# Creating a second y-axis for Child Mortality
ax2 = ax1.twinx()
sns.lineplot(data=benchmarking_data, x='country', y='child_mort', marker='o',
            color='red', linewidth=3, ax=ax2, label='Child Mortality')
```

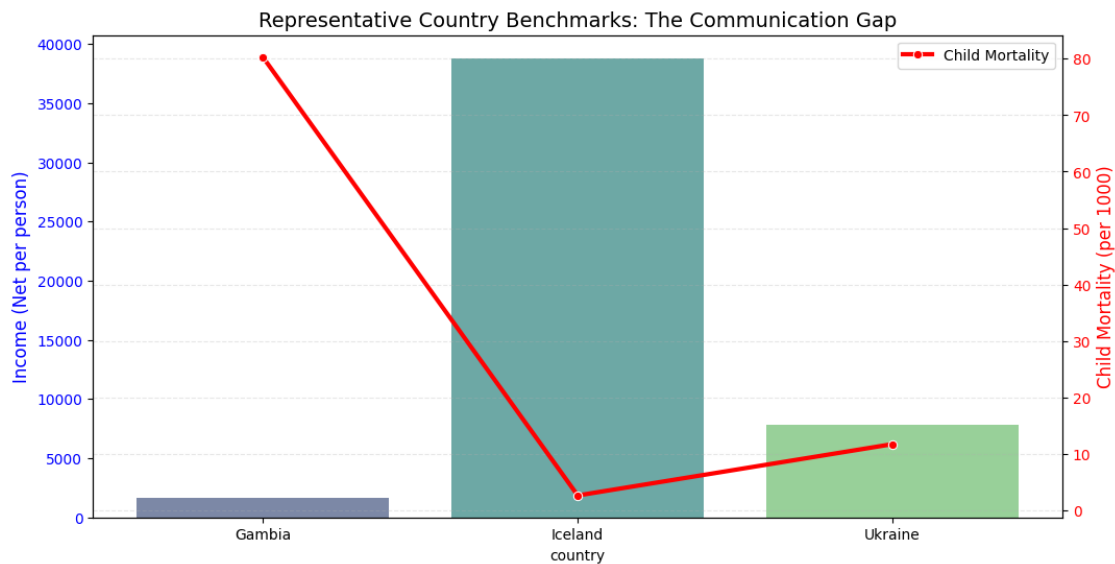
```

ax2.set_ylabel('Child Mortality (per 1000)', color='red', fontsize=12)
ax2.tick_params(axis='y', labelcolor='red')

plt.title('Representative Country Benchmarks: The Communication Gap',
         ↪ fontsize=14)
plt.grid(axis='y', linestyle='--', alpha=0.3)
plt.show()

display(benchmarking_data[['country', 'cluster_id_pca', 'income', 'child_mort',
         ↪ 'life_expec']])

```



0.1.13 Visualizing the Impact: Health Spending vs. Child Mortality

To understand the efficiency of resource allocation, we visualize the correlation between health spending (as a % of GDP) and child mortality. This helps determine if increased spending directly translates to better survival rates across different development tiers.

```

[59]: plt.figure(figsize=(12, 8))

# Using a regression plot to see the trend within each cluster
sns.lmplot(data=country_data, x='health', y='child_mort', hue='cluster_id_pca',
           palette='viridis', height=7, aspect=1.2, scatter_kws={'alpha':0.6})

```

```
plt.title('Correlation: Health Spending vs. Child Mortality by Cluster',
         ↪fontsize=14)
plt.xlabel('Health Spending (% of GDP)', fontsize=12)
plt.ylabel('Child Mortality (per 1000 births)', fontsize=12)
plt.grid(True, linestyle='--', alpha=0.5)

# Adding descriptive labels for clarity
plt.legend(title='Clusters', labels=['At-Risk', 'Developed', 'Developing'])
plt.show()
```

<Figure size 1200x800 with 0 Axes>



The regression plot reveals critical insights into health spending efficiency across the clusters:

1. **At-Risk Cluster (Cluster 0):** The regression line is nearly flat or shows a very high intercept. This suggests that in these nations, simply increasing the percentage of GDP spent on health hasn't yet translated into significant mortality reduction. This points to a need for structural healthcare reform and basic infrastructure rather than just capital injection.
2. **Developing Cluster (Cluster 2):** Shows a clearer negative slope, indicating a higher 'return on investment' for health spending compared to the At-Risk group.
3. **Developed Cluster (Cluster 1):** The line is flat but at a near-zero mortality rate, representing a 'maintenance' phase where high spending sustains excellent outcomes.

0.1.14 Strategic Rationale: The Business Purpose of Representative Countries

Identifying the nation closest to each cluster’s centroid—our **Representative Countries**—serves three primary strategic functions for policy communication:

1. **Humanizing the Data (Personas):** Stakeholders often struggle with abstract concepts like ‘Cluster 0’ or ‘PC1 scores.’ By using **Gambia**, **Ukraine**, and **Iceland**, we transform mathematical segments into relatable “personas.” This makes the socioeconomic reality of each group concrete and understandable for non-technical decision-makers.
2. **Facilitating the “Communication Gap”:** As seen in the benchmark plot above, the disparity between Gambia and Iceland is massive. Visualizing these specific anchors justifies why a **one-size-fits-all policy is ineffective**. It provides a clear visual argument for tiered resource allocation based on the specific needs of the representative profile.
3. **Baseline & Pilot Modeling:** These representative nations serve as the **ideal pilot cases**. A policy successfully implemented in Gambia provides a statistically sound “proof of concept” for the other 51 countries in the At-Risk cluster, as Gambia represents the median state of that entire group.